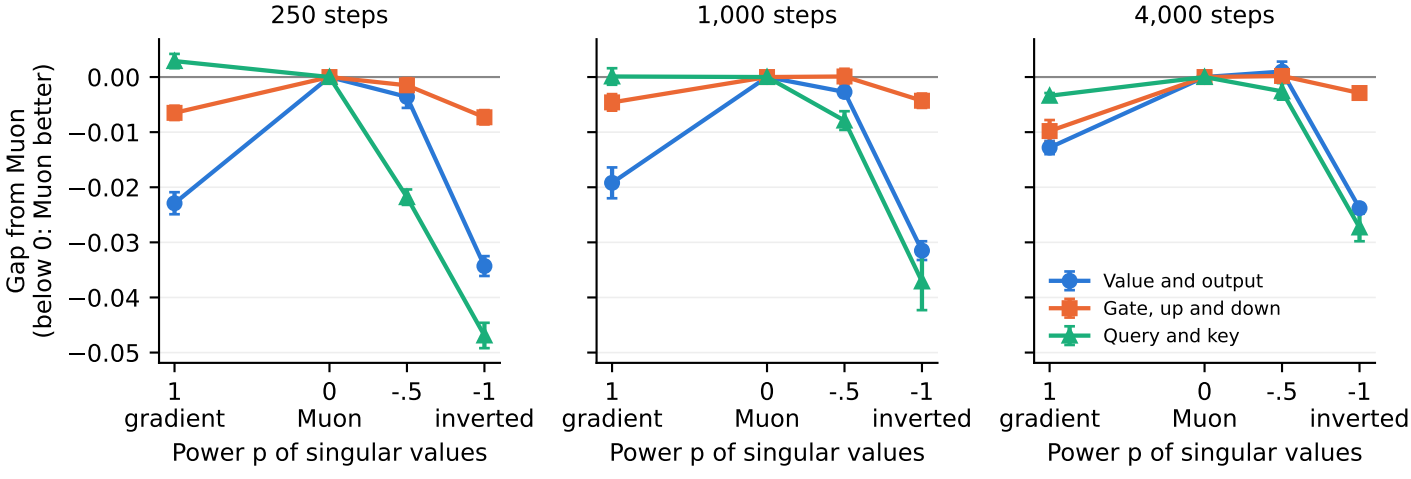
250 steps
1,000 steps
4,000 steps
Gap from Muon
(below 0: Muon better)
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
1
gradient
0
Muon
-.5
-1
inverted
Power p of singular values
Value and output
Gate, up and down
Query and key